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Exercise Sheet 3

This is not the official exercise sheet from the course.
More difficult exercises are denoted by ★ to ★★★

> If you can solve it, it is an exercise;
> otherwise it's a research problem.
>
> Richard Bellman

Exercise 1.
For each of the following sets A and B, decide which of $A \in B, A \subseteq B, A \ni B, A \supseteq B$ is true.

1. $A = \{\{1,2\},3\}$, $B = \{1,2\}$
2. $A = \mathbb{N}$, $B = \mathbb{R}$
3. $A = \{\{\emptyset\},\emptyset\}$, $B = \{\emptyset,\{\emptyset\}\}$
4. $A = \{x \in \mathbb{R} : x^2 - 4 = 0\}$, $B = \{x \in \mathbb{Z} : |x| \leq 2\}$
5. $A = \{\emptyset,\{\emptyset\}\}$, $B = \emptyset$
6. $A = \{\emptyset,\{\emptyset\},\{\{\emptyset\}\},\{\emptyset,\{\emptyset\}\}\}, B = \{\emptyset,\{\emptyset\}\}$
7. $A = \emptyset$, $B = \mathbb{N}$
8. $A = \{\emptyset\}$, $B = \{\emptyset,\{\emptyset\}\}$
9. $A = \{\emptyset,\{\emptyset\}\}$, $B = \{\{\emptyset\}\}$
10. $A = \mathcal{P}(\emptyset)$, $B = \mathcal{P}(\mathcal{P}(\emptyset))$
11. $A = \{\emptyset\}$, $B = \mathcal{P}(\{\emptyset\})$
12. $A = \{\{\emptyset,\{\emptyset\}\}\}$, $B = \{\emptyset,\{\emptyset\},\{\emptyset,\{\emptyset\}\}\}$
13. $A = \{\emptyset \setminus \{\emptyset\}\}$, $B = \emptyset$
14. $A = \mathcal{P}(\emptyset) \setminus \emptyset$, $B = \{\emptyset\} \setminus \{\emptyset\}$

Exercise 2.
Let $|A| = n$ and $|B| = m$. Give the cardinalities of the following sets in terms of n, m.

1. $|\mathcal{P}(\mathcal{P}(\mathcal{P}(A)))|$
2. $|\mathcal{P}(A \times B)|$
3. $|\mathcal{P}(A) \times \{\emptyset\}|$
4. $|\mathcal{P}(\mathcal{P}(\mathcal{P}(A \times \emptyset)))|$
5. $|\mathcal{P}(\mathcal{P}(A)) \times \mathcal{P}(\mathcal{P}(B))|$
6. $|(\mathcal{P}(A) \times B) \setminus \emptyset|$

Exercise 3.
Let A, B be arbitrary sets. Prove or disprove each of the following statements.

1. $\mathcal{P}(A \times B) = \mathcal{P}(A) \times \mathcal{P}(B)$
2. $\mathcal{P}(A \setminus B) = \mathcal{P}(A) \setminus \mathcal{P}(B)$
3. If $\mathcal{P}(A) \subseteq \mathcal{P}(B)$, then $A \subseteq B$

Exercise 4.
Let A, B be finite sets. Prove or disprove each of the following statements.

1. $|\mathcal{P}(A \cup B)| = |\mathcal{P}(A)| + |\mathcal{P}(B)| - |\mathcal{P}(A \cap B)|$
2. $|\mathcal{P}(A \times B)| = 2^{|A| \cdot |B|}$
3. If $|\mathcal{P}(A)| = |\mathcal{P}(B)|$, then $A = B$

★ **Exercise 5.**

Let Let $A_n = \{n\}$ for each $n \in \mathbb{N}$ be a sequence of sets and $\mathbb{P}$ the set of all prime numbers. Compute

1. $\bigcup_{n=0}^{\infty} A_n$
2. $\bigcap_{n=0}^{\infty} A_n$
3. $\bigcap_{n=0}^{\infty} A_{3n}$
4. $\left(\bigcup_{n\in\mathbb{P}} A_n\right) \setminus \left(\bigcup_{n\in\mathbb{N}\setminus\mathbb{P}} A_n\right)$
5. $\bigcap_{n=0}^{\infty} (A_n \cup A_{n+1})$
6. $\bigcup_{n=1}^{\infty} (A_n \setminus A_{n-1})$